Cornerstones or Stumbling Blocks? Deciphering the Role of Recalcitrant Tokens in On-Policy Distillation

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Abstract

On-Policy Distillation (OPD) has emerged as a principled paradigm for transferring capabilities from large teacher models to compact student models, mitigating the distribution mismatch that plagues off-policy approaches. However, a systematic understanding of *which tokens* drive or hinder the distillation process remains lacking. We introduce the concept of **Recalcitrant Tokens** (or "Rock Tokens")—tokens that maintain persistently high distillation loss both before and after the OPD process—and propose a causal investigation into their functional role. We hypothesize that Rock Tokens bifurcate into two categories: **Pillar Tokens**, whose presence is structurally essential and whose removal causes training collapse, and **Stumbling-Block Tokens**, whose removal accelerates learning and improves final performance. Our methodology combines token-level loss tracking across OPD training, constrained decoding experiments that prohibit recalcitrant token generation, and cross-model transfer analyses. Using Qwen3-30B-Instruct as teacher and Qwen3-4B-Instruct and Llama-3.2-3B as students, we aim to establish a causal taxonomy of token-level contributions to OPD, with implications for adaptive loss weighting, curriculum design, and efficient distillation.

1 Introduction

Knowledge distillation [Hinton et al., 2015] has evolved from a niche compression technique into a central pillar of the modern large language model (LLM) development pipeline. The release of DeepSeek-R1 [DeepSeek-AI et al., 2025], which transferred complex chain-of-thought reasoning from a 671-billion-parameter mixture-of-experts teacher into dense students ranging from 1.5B to 70B parameters, exemplifies how distillation now serves as a general-purpose capability transfer engine rather than a mere size-reduction tool.

Traditional *off-policy* distillation trains the student on fixed teacher-generated sequences, inducing a well-documented distribution mismatch: the student never learns to recover from its own errors because training data reflects the teacher's policy, not the student's [Agarwal et al., 2024]. *On-Policy Distillation* (OPD) addresses this by letting the student sample sequences from its own evolving policy and soliciting token-level teacher feedback on these self-generated trajectories [Gu et al., 2024, Agarwal et al., 2024]. Recent work has demonstrated that OPD achieves faster and more effective distillation than off-policy alternatives across reasoning, instruction-following, and multi-task settings [Yang et al., 2025, Ye et al., 2025, Zhao et al., 2026].

Despite these advances, the token-level dynamics of OPD remain poorly understood. Recent studies on reinforcement learning with verifiable rewards (RLVR) have revealed that only a small minority of high-entropy "forking tokens" drive reasoning improvements, while the majority of low-entropy tokens contribute minimally [Wang et al., 2025]. Similarly, adaptive distillation methods such as AdaKD [Xie et al., 2025] have shown that focusing the distillation loss on the most difficult tokens can improve performance and efficiency. These findings suggest that tokens are *not* created equal in the learning process.

However, a crucial gap remains. Prior work has studied tokens that are *eventually learned* (high initial loss that decreases) or tokens that are *always easy* (consistently low loss). An unexplored category is what we term **Recalcitrant Tokens** or **Rock Tokens**: tokens that maintain persistently high loss throughout the entire OPD process, resisting the student's learning signal from start to finish. The central question of this proposal is:

Are Recalcitrant Tokens essential structural components ("Pillars") whose presence anchors the student's learned distribution, or are they detrimental noise ("Stumbling Blocks") that waste gradient capacity and impede convergence?

This question has immediate practical implications. If Rock Tokens are predominantly Stumbling Blocks, then detecting and removing them could yield faster convergence and better final performance—a form of token-level curriculum learning. If they are predominantly Pillars, then current practices of uniform loss weighting are inadvertently correct, and targeted up-weighting of these tokens might further improve distillation. If, as we suspect, the category is heterogeneous, then developing principled methods to distinguish Pillars from Stumbling Blocks becomes a prerequisite for next-generation OPD algorithms.

Contributions. We propose:

1. A formal definition and identification methodology for Recalcitrant Tokens in OPD.
2. A causal intervention framework that prohibits recalcitrant token generation to assess their functional impact on reasoning and format adherence.
3. Cross-model and cross-family experiments to determine whether Rock Token identity is model-specific or reflects intrinsic properties of the data or task.
4. A preliminary investigation into whether the removal of Stumbling Blocks triggers the emergence of new Pillar Tokens, suggesting a dynamic structural reorganization of the student's token distribution.

2 Related Work

2.1 On-Policy Distillation for LLMs

On-policy distillation addresses the train-test distribution mismatch inherent in standard knowledge distillation by training the student on its own generated sequences. MiniLLM [Gu et al., 2024] proposed replacing the standard forward KL divergence with reverse KL divergence, deriving an on-policy optimization objective that prevents the student from overestimating low-probability regions of the teacher distribution. Generalized Knowledge Distillation (GKD) [Agarwal et al., 2024] further demonstrated that training on student-generated outputs with teacher feedback can be seamlessly integrated with RLHF.

Recent work has expanded OPD in several directions. Cross-family distillation has been explored, enabling knowledge transfer between architecturally distinct model families [Patiño et al., 2025]. Black-box OPD methods such as Generative Adversarial Distillation (GAD) [Ye et al., 2025] frame the student as a generator and train a discriminator to distinguish student from teacher text, enabling distillation without access to teacher logits. Self-Distilled Reasoner [Zhao et al., 2026] showed that a model can serve as its own teacher by conditioning on privileged answer information. The recent survey by Survey [2026] provides a comprehensive taxonomy, noting that methods from the knowledge distillation, RLHF, and imitation learning communities often address the same underlying problem with different formalisms.

A parallel line of work connects OPD to continual learning. On-policy learning has been shown to forget less than off-policy supervised fine-tuning [Shenfeld et al., 2025], and self-distillation has been proposed as a practical path to continual learning from demonstrations [Thinking Machines Lab, 2025]. These findings are particularly relevant to our work, as they suggest that the tokens the student *fails to learn* (our Rock Tokens) may play a role in determining what knowledge is preserved or lost.

2.2 Token-Level Analysis in LLM Training

The insight that individual tokens contribute heterogeneously to learning has gained significant traction. Wang et al. [2025] demonstrated that in RLVR, approximately 20% of tokens exhibit high entropy and function as "forking tokens" that steer reasoning into diverse pathways. Restricting policy gradient updates to only these forking tokens maintained or improved performance compared to full-gradient updates, while training exclusively on the 80% lowest-entropy tokens degraded performance markedly. This finding was published at NeurIPS 2025 and established a direct link between token-level entropy and learning efficacy.

In the distillation literature, AdaKD [Xie et al., 2025] introduced Loss-Driven Adaptive Token Focusing, which restricts the distillation loss to the most difficult tokens in each training phase, combined with Inverse

Difficulty Temperature Scaling. The framework achieves improved generalization stability by synchronizing token selection and temperature modulation. Similarly, AdaSPEC [Hu et al., 2025] proposed token loss filtering based on loss-gap ranking. The causal attention distillation work of Guo et al. [2025] explicitly identifies and removes "confounding tokens" based on sensitivity analysis of teacher attention, distilling only on causally relevant context positions.

These works focus on tokens that vary in *difficulty during training*. Our contribution is orthogonal: we study tokens that are *persistently* difficult—never learned despite the entire OPD process—and ask whether this persistence is informative of their causal role.

2.3 Catastrophic Forgetting and Knowledge Preservation

Catastrophic forgetting [Kirkpatrick et al., 2017] during fine-tuning remains a central challenge. Learning without Forgetting (LwF) [Li & Hoiem, 2018] pioneered the use of knowledge distillation from a prior checkpoint to preserve capabilities. Recent work has shown that RL-based training naturally mitigates forgetting compared to SFT [Shenfeld et al., 2025], potentially because the KL divergence regularization in RL acts as implicit knowledge distillation. The instruction vector hypothesis [Li et al., 2024] provides mechanistic insight, showing that fine-tuning suppresses instruction-related computation graphs rather than erasing knowledge outright.

Our work connects to this literature by asking whether Rock Tokens represent knowledge that *should not* be transferred (Stumbling Blocks, analogous to noise that causes forgetting when the student tries to fit it) or knowledge that *must* be transferred (Pillars, analogous to the instruction vectors whose preservation prevents catastrophic failure).

3 Problem Formulation

3.1 On-Policy Distillation Setup

Let π_T denote the teacher model and π_θ the student model parameterized by θ. In OPD, at each training iteration, the student generates a sequence $\mathbf{y} = (y_1, y_2, \ldots, y_L)$ conditioned on a prompt $\mathbf{x}$ by sampling from its own policy: $y_t \sim \pi_\theta(\cdot \mid \mathbf{x}, y_{<t})$. The teacher then provides a supervisory signal, typically in the form of its full output distribution $\pi_T(\cdot \mid \mathbf{x}, y_{<t})$ at each position t. The student is updated to minimize a divergence measure:

$$\mathcal{L}_{\mathrm{OPD}}(\theta) = \mathbb{E}_{\mathbf{x}\sim\mathcal{D},\, \mathbf{y}\sim\pi_\theta(\cdot|\mathbf{x})} \left[\sum_{t=1}^{L} D_{\mathrm{KL}} \left(\pi_T(\cdot \mid \mathbf{x}, y_{<t}) \,\|\, \pi_\theta(\cdot \mid \mathbf{x}, y_{<t}) \right) \right] \tag{1}$$

where $\mathcal{D}$ is the prompt distribution. Following MiniLLM [Gu et al., 2024], we may alternatively use reverse KL divergence to prevent mode-covering behavior.

3.2 Defining Recalcitrant Tokens

We define the *token-level distillation loss* at position t in sequence $\mathbf{y}$ as:

$$\ell_t(\theta) = D_{\mathrm{KL}} \left(\pi_T(\cdot \mid \mathbf{x}, y_{<t}) \,\|\, \pi_\theta(\cdot \mid \mathbf{x}, y_{<t}) \right) \tag{2}$$

Let θ_0 denote the student parameters before OPD and θ^* the parameters after convergence. A token y_t in context $(\mathbf{x}, y_{<t})$ is **Recalcitrant** if:

$$\ell_t(\theta_0) > \tau \quad \text{and} \quad \ell_t(\theta^*) > \tau \tag{3}$$

for a threshold τ. In practice, we set τ at the p-th percentile (e.g., $p = 90$) of the loss distribution at initialization, and require that the loss remains above this threshold at convergence. This identifies tokens where the student-teacher divergence is high and *stays* high.

3.3 Taxonomy of Token Behavior

By tracking $\ell_t(\theta_0)$ and $\ell_t(\theta^*)$, we can partition tokens into four quadrants (Figure 1):

1. **Easy Tokens**: Low initial loss, low final loss. Already aligned; minimal learning needed.
2. **Learned Tokens**: High initial loss, low final loss. Successfully transferred during OPD.
3. **Recalcitrant Tokens (Rock Tokens)**: High initial loss, high final loss. The focus of this work.
4. **Regressed Tokens**: Low initial loss, high final loss. Potentially indicating catastrophic forgetting.

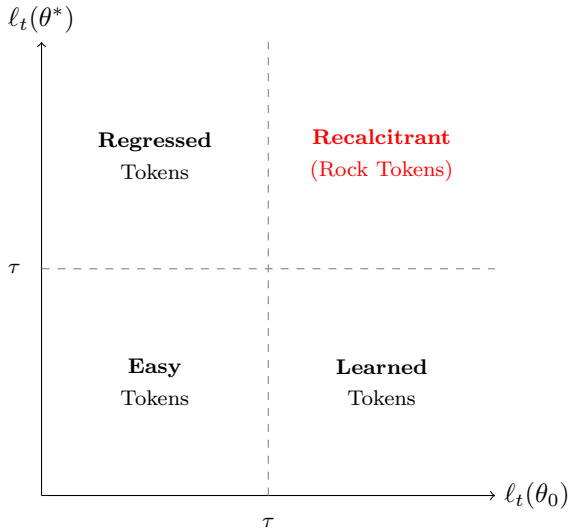

Figure 1: Taxonomy of token behavior during OPD based on initial and final distillation loss.

4 Proposed Methodology

Our experimental plan proceeds in three stages, each designed to address a distinct research question.

4.1 Step 1: Establishing the OPD Framework and Identifying Rock Tokens

Experiment 1: OPD Baseline (Lead: Zhao). We establish the OPD experimental framework using:

- **Teacher**: Qwen3-30B-Instruct
- **Students**: Qwen3-4B-Instruct (same family), Llama-3.2-3B (cross-family)
- **Datasets**: A diverse mixture of mathematical reasoning (GSM8K, MATH), code generation (HumanEval, MBPP), and general instruction-following (Tulu3 prompts)

We implement white-box OPD following the GKD framework [Agarwal et al., 2024], where the student samples on-policy and receives token-level KL feedback from the teacher. Training proceeds for N iterations with periodic checkpointing. At each checkpoint k, we record the per-token loss $\ell_t(\theta_k)$ on a fixed evaluation set, enabling us to construct complete loss trajectories for every token position.

Experiment 2: Rock Token Identification (Lead: Chao). Using the loss trajectories from Experiment 1, we identify Rock Tokens according to the criterion in Equation (3). We additionally compute:

- **Token entropy** $H_t = -\sum_v \pi_\theta(v \mid \mathbf{x}, y_{<t}) \log \pi_\theta(v \mid \mathbf{x}, y_{<t})$ to relate Rock Tokens to the forking token framework of Wang et al. [2025].
- **Semantic analysis**: Categorize Rock Tokens by type (punctuation, function words, content words, reasoning connectives, mathematical symbols).
- **Positional analysis**: Examine whether Rock Tokens cluster at sequence boundaries, reasoning transitions, or uniformly.

- **Cross-model overlap**: Compute the Jaccard similarity of Rock Token sets between the two student models to determine whether recalcitrance is model-dependent or data-intrinsic.

4.2 Step 2: Causal Impact Assessment

Method: Constrained Decoding. To establish the *causal* role of Rock Tokens (beyond correlational analysis), we employ an interventionist approach: we explicitly prohibit the student model from generating identified Rock Tokens during inference by setting their logit to $-\infty$ before the softmax operation:

$$\pi_\theta^{\text{constrained}}(v \mid \mathbf{x}, y_{<t}) = \begin{cases} 0 & \text{if } v \in \mathcal{R} \\ \frac{\pi_\theta(v|\mathbf{x}, y_{<t})}{\sum_{v' \notin \mathcal{R}} \pi_\theta(v'|\mathbf{x}, y_{<t})} & \text{otherwise} \end{cases} \tag{4}$$

where $\mathcal{R}$ is the identified set of Recalcitrant Token types.

Evaluation Metrics. We evaluate the impact of Rock Token removal along two dimensions:

1. **Reasoning Performance**: Accuracy on GSM8K, MATH, AIME benchmarks, and pass@1 on HumanEval/MBPP.
2. **Output Format**: IF-Eval scores, response length distribution, structural coherence (measured via format-specific regex matching for mathematical answers).

Expected Outcomes and Interpretation.

- **If Rock Tokens are Pillars**: Removal causes degradation in both reasoning accuracy and format adherence. The model may produce incoherent outputs, lose the ability to structure multi-step reasoning, or collapse to degenerate repetitive sequences. This would indicate that these tokens, while never fully learned, serve as essential anchoring points in the student's generation process.
- **If Rock Tokens are Stumbling Blocks**: Removal improves or maintains reasoning accuracy while potentially improving output coherence. This would suggest that the student wastes gradient capacity on tokens it cannot learn, and that removing them frees the model to allocate resources more effectively.
- **Mixed regime** (most likely): Some Rock Tokens are Pillars and others are Stumbling Blocks, necessitating finer-grained classification methods.

4.3 Step 3: Dynamic Token Ecology (Proposed Future Work)

Experiment 3: Cross-Model Transfer and Pillar Emergence. Building on the findings of Steps 1–2, we investigate three questions:

1. **Cross-model token removal**: Are Stumbling Blocks identified in Qwen3-4B also Stumbling Blocks for Llama-3.2? This tests whether the Pillar/Stumbling Block distinction is a property of the data/task or of the specific student architecture.
2. **Stumbling Block detection**: Can we develop an efficient classifier that predicts at initialization which Rock Tokens will be Stumbling Blocks, enabling early removal? Candidate features include token entropy, teacher-student gradient alignment, and token frequency statistics.
3. **Pillar emergence**: After removing identified Stumbling Blocks and re-running OPD, do new Pillar Tokens emerge at different positions? If so, this suggests that the student's learning process involves a dynamic structural reorganization, where the "load-bearing" role previously filled by certain tokens is redistributed. This would have profound implications for understanding the topology of the student's learned representation.

5 Experimental Design

5.1 Models and Infrastructure

The inclusion of both same-family (Qwen3→Qwen3) and cross-family (Qwen3→Llama) distillation allows us to disentangle the effect of shared vocabulary and architecture from the intrinsic difficulty of certain tokens.

Table 1: Model configurations for the proposed experiments.

Role	Model	Parameters	Family
Teacher	Qwen3-30B-Instruct	30B	Qwen3
Student 1	Qwen3-4B-Instruct	4B	Qwen3
Student 2	Llama-3.2-3B	3B	Llama

5.2 Training Protocol

We follow the GKD [Agarwal et al., 2024] protocol with the following hyperparameters: learning rate 1×10^{-5} with cosine schedule, batch size 128, KL divergence loss (forward and reverse variants), and training for 10K steps. We checkpoint every 500 steps and compute full token-level loss statistics on a held-out evaluation set of 5K prompts at each checkpoint.

5.3 Evaluation Benchmarks

Table 2: Evaluation benchmarks and metrics.

Benchmark	Domain	Metric
GSM8K	Math reasoning	Accuracy
MATH	Advanced math	Accuracy
AIME 2024/2025	Competition math	Score
HumanEval	Code generation	Pass@1
MBPP	Code generation	Pass@1
IF-Eval	Instruction following	Accuracy

6 Theoretical Motivation

We provide informal theoretical grounding for why Rock Tokens should exist and why their role may be heterogeneous.

Capacity gap argument. The student model has strictly fewer parameters than the teacher, implying a representational capacity gap. Not all teacher behaviors can be faithfully represented by the student. Tokens that lie in regions of the output space where the capacity gap is most severe will naturally remain high-loss. If these tokens correspond to behaviors that are *essential* for downstream task performance (e.g., reasoning connectives, step-transition markers), they are Pillars. If they correspond to idiosyncratic teacher behaviors that do not contribute to task performance (e.g., stylistic preferences, rare vocabulary choices), they are Stumbling Blocks.

Connection to mode-covering vs. mode-seeking. Forward KL (mode-covering) forces the student to assign probability mass everywhere the teacher does, including on tokens it cannot model well—potentially creating Stumbling Blocks. Reverse KL (mode-seeking) allows the student to ignore low-probability teacher modes—potentially losing Pillars. The choice of divergence measure directly influences which tokens become Recalcitrant and whether they function as Pillars or Stumbling Blocks.

Information-theoretic perspective. Rock Tokens may correspond to positions where the teacher distribution has high entropy (many plausible next tokens) but the student distribution has collapsed to a different mode. In this case, the KL divergence remains high not because the student assigns low probability to the correct token, but because the full distributional shape diverges. This connects our framework to the

forking token analysis of Wang et al. [2025], where high-entropy positions are the critical decision points for reasoning.

7 Broader Impact and Implications

Efficient distillation. If Stumbling Blocks can be reliably identified early in training, adaptive loss weighting that down-weights these tokens could reduce the effective training time and computational cost of OPD. Given that maintaining a large teacher model in memory alongside the student demands substantial engineering and compute [Ko et al., 2024], any reduction in training iterations translates to significant resource savings.

Curriculum design. The Pillar/Stumbling Block taxonomy provides a principled basis for token-level curriculum learning in distillation: begin with Easy and Learned tokens, gradually introduce Pillar tokens with increased weight, and suppress Stumbling Blocks throughout.

Understanding distillation failures. When distillation fails to transfer a capability (e.g., multi-step reasoning), Rock Token analysis can provide diagnostic insight into *which specific tokens* the student fails to learn, enabling targeted interventions.

Connection to model interpretability. The identity of Pillar Tokens—tokens whose removal collapses generation—provides a window into the structural dependencies of the student's learned distribution, complementing existing interpretability methods.

8 Timeline and Resource Requirements

Table 3: Projected timeline.

Phase	**Duration**	**Activities**
Phase 1	Weeks 1–4	OPD baseline, infrastructure setup
Phase 2	Weeks 3–6	Rock Token identification and analysis
Phase 3	Weeks 5–10	Causal intervention experiments
Phase 4	Weeks 8–12	Cross-model transfer, Pillar emergence
Phase 5	Weeks 11–14	Paper writing, ablations

Compute requirements: approximately 8×A100 (80GB) GPUs for 4 weeks, primarily for OPD training with full teacher-in-the-loop and evaluation sweeps.

9 Conclusion

We propose a systematic investigation into Recalcitrant Tokens—the persistently high-loss tokens in On-Policy Distillation—and their causal role in the distillation process. By combining token-level loss tracking, constrained decoding interventions, and cross-model analyses, we aim to establish whether these tokens are essential structural Pillars or detrimental Stumbling Blocks. The findings will inform the design of more efficient and principled OPD algorithms, contributing to the broader goal of scalable, effective knowledge transfer in the era of large language models.

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